which are called the **Cartesian equations or symmetric equations** of a straight line passing through the point (x_1, y_1, z_1) and parallel to a vector with direction ratios b_1, b_2, b_3.

Remark

(i) Every point on the line (5) is of the form $(x_1 + tb_1, y_1 + tb_2, z_1 + tb_3)$, where $t \in \mathbb{R}$.

(ii) Since the direction cosines of a line are proportional to direction ratios of the line, if l, m, n are the direction cosines of the line, then the Cartesian equations of the line are

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}.$$

(iii) In (5), if any one or two of b_1, b_2, b_3 are zero, it does not mean that we are dividing by zero. But it means that the corresponding numerator is zero. For instance, If $b_1 \neq 0$, $b_2 \neq 0$ and $b_3 = 0$, then $\frac{x - x_1}{b_1} = \frac{y - y_1}{b_2} = \frac{z - z_1}{0}$ should be written as $\frac{x - x_1}{b_1} = \frac{y - y_1}{b_2}, z - z_1 = 0$.

(iv) We know that the direction cosines of x-axis are $1, 0, 0$. Therefore, the equations of x-axis are

$$\frac{x - 0}{1} = \frac{y - 0}{0} = \frac{z - 0}{0} \text{ or } x = t, y = 0, z = 0, \text{ where } t \in \mathbb{R}.$$

Similarly the equations of y-axis and z-axis are given by $\frac{x - 0}{0} = \frac{y - 0}{1} = \frac{z - 0}{0}$ and $\frac{x - 0}{0} = \frac{y - 0}{0} = \frac{z - 0}{1}$ respectively.

6.7.3 Straight Line passing through two given points

(a) Parametric form of vector equation

Theorem 6.12

The parametric form of vector equation of a line passing through two given points whose position vectors are $\vec{a}$ and $\vec{b}$ respectively is $\vec{r} = \vec{a} + t(\vec{b} - \vec{a}), t \in \mathbb{R}$.

(b) Non-parametric form of vector equation

The above equation can be written equivalently in non-parametric form of vector equation as

$$(\vec{r} - \vec{a}) \times (\vec{b} - \vec{a}) = \vec{0}$$

(c) Cartesian form of equation

Suppose P is (x, y, z), A is (x_1, y_1, z_1) and B is (x_2, y_2, z_2). Then substituting $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $\vec{a} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$ and $\vec{b} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$ in theorem 6.12 and comparing the coefficients of $\hat{i}, \hat{j}, \hat{k}$, we get $x - x_1 = t(x_2 - x_1), y - y_1 = t(y_2 - y_1), z - z_1 = t(z_2 - z_1)$ and so the Cartesian equations of a line passing through two given points (x_1, y_1, z_1) and (x_2, y_2, z_2) are given by

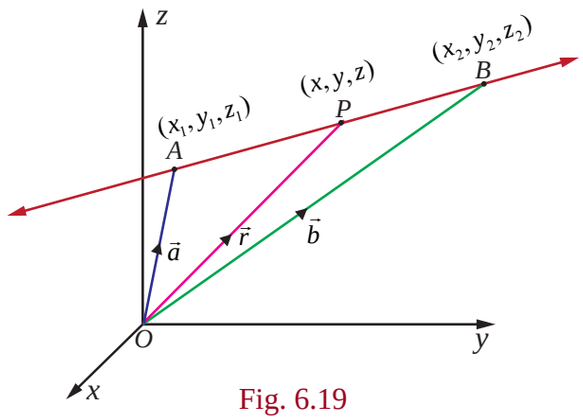


Fig. 6.19

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